

TREVOR HYDE

ASSISTANT PROFESSOR OF MATHEMATICS AND STATISTICS

Vassar College

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RESEARCH

I work in arithmetic dynamics and number theory, with excursions into algebraic geometry, combinatorics, representation theory, and topology. Recurring themes in my work include rational points in orbits, preperiodic points, and profinite iterated monodromy groups; the statistics of polynomial factorization over finite fields and their connection to the cohomology of configuration spaces; and explicit Galois theory. My recent work focuses on the arithmetic of absolute geometry.

COMPUTATIONAL AND FORMAL METHODS

- **Lean / formal verification.** Designed and taught *Formalizing Mathematics* (Math 301, senior seminar, Spring 2025), an intensive course on the Lean 4 proof assistant ([course repository](#)). Self-taught in Lean since 2024.
- **AI tools in research.** Exploring the use of AI tools in my research since late 2022; regular user of Claude, ChatGPT, and Gemini. Interested in alignment and safety of AI.
- **Software projects.** AI-assisted coding projects on [GitHub](#), including [hwgenie](#) (bespoke course management software), [Row Reducer](#) (an interactive row-reduction practice tool with exact arithmetic), [Portalz](#) (visualization tool for exploring branched coverings of 3-manifolds), and [Ricochet Solitaire](#) (a solitaire version of the board game *Ricochet Robots*).
- **Computer algebra and programming.** SageMath and Python, for research computation (e.g. the [Sage notebook supplement](#) to publication 7 below) and for exploratory problem sets in number theory and linear algebra courses.

EDUCATION

Ph.D. , Mathematics	University of Michigan	2019
Thesis: <i>Polynomial Statistics, Necklace Polynomials, and the Arithmetic Dynamical Mordell–Lang Conjecture</i>		
Advisors: Jeffrey Lagarias and Michael Zieve		
B.A. , Mathematics, <i>summa cum laude</i> , Phi Beta Kappa	Amherst College	2012

APPOINTMENTS

Assistant Professor of Mathematics and Statistics	Vassar College	2023–present
NSF Mathematical Sciences Postdoctoral Research Fellow	University of Chicago	2020–2023
L. E. Dickson Instructor	University of Chicago	2019–2023
Moss Quantitative Fellow	Amherst College	2012–2013

GRANTS, FELLOWSHIPS, AND AWARDS

AMS–Simons Research Enhancement Grant for PUI Faculty	AMS / Simons Foundation	2026–2029
NSF Mathematical Sciences Postdoctoral Research Fellowship (DMS-2002176)	National Science Foundation	2020–2023
Swogger Award for Exemplary Classroom Teaching	University of Chicago	2022
Rackham Predoctoral Fellowship	University of Michigan	2018
Wirt and Mary Cornwell Prize	University of Michigan	2018
Rackham Outstanding Graduate Student Instructor Award	University of Michigan	2018

Department of Mathematics Outstanding Teaching Award	University of Michigan	2017
Robert Breuch Thesis Prize	Amherst College	2012
Post-Baccalaureate Fellowship	Amherst College	2012

PUBLICATIONS

* denotes an undergraduate coauthor at the time of the work.

Preprints

1. The arithmetic of semirings part I: ideals, with J. Chen*, D. Laurens*, J. Piermarini*, and H. Simons*, submitted (2026), [arXiv:2607.10778](#).
2. Rational orbits under correspondences, submitted (2026), [arXiv:2605.04391](#).

Refereed journal articles

3. Profinite iterated monodromy groups of unicritical polynomials, with O. Adams, to appear in *Algebra Number Theory* (2026), [arXiv:2504.13028](#).
4. Polynomials with many rational preperiodic points, with J. R. Doyle, *Trans. Amer. Math. Soc.* **378** (2025), no. 8, 5981–6011, [arXiv:2201.11707](#).
5. Cyclotomic factors of necklace polynomials, *Acta Arith.* **204** (2022), 287–316.
6. Dynatomic polynomials, necklace operators, and universal relations for dynamical units, with J. R. Doyle and P. Fili, *New York J. Math.* **28** (2022), 534–556, [arXiv:2108.09333](#).
7. Dynamical moduli spaces and polynomial endomorphisms of configurations, with T. Blum*, J. R. Doyle, C. Kelln*, H. Talbott*, and M. Weinreich, *Arnold Math. J.* **8** (2022), 285–317.
8. Euler characteristic of the space of real multivariate irreducible polynomials, *Proc. Amer. Math. Soc.* **150** (2022), 2331–2343.
9. Density of periodic points for Lattès maps over finite fields, with Z. Bell*, J. Camero*, K. Cho*, C. Lu*, R. Miller, B. Thompson, and E. Zhu*, *J. Number Theory* **238** (2022), 951–966.
10. Polynomial factorization statistics and point configurations in $\mathbb{R}^3$, *Int. Math. Res. Not. IMRN* **2020**, no. 24, 10154–10179.
11. Liminal reciprocity and factorization statistics, *Algebr. Comb.* **2** (2019), no. 4, 521–539.
12. Polynomial splitting measures and cohomology of the pure braid group, with J. C. Lagarias, *Arnold Math. J.* **3** (2017), no. 2, 219–249.
13. Gauss’ hidden menagerie: from cyclotomy to supercharacters, with S. R. Garcia and B. Lutz, *Notices Amer. Math. Soc.* **62** (2015), no. 8, 878–888.
14. Galois theory of the lemniscate, with D. A. Cox, *J. Number Theory* **135** (2014), 43–59.
15. Small dynamical heights for quadratic polynomials and rational functions, with R. Benedetto, R. Chen, Y. Kovacheva, and C. White, *Exp. Math.* **23** (2014), 433–447.
16. A Wallis product on clovers, *Amer. Math. Monthly* **121** (2014), no. 3, 237–243.
17. Pre-images in quadratic dynamical systems, with B. Hutz and B. Krause, *Involve* **4** (2011), no. 4, 343–363.

Conference proceedings and thesis

18. Factorization statistics and the twisted Grothendieck–Lefschetz formula, extended abstract, *Sém. Lothar. Combin.* **80B** (2018), Art. 6 (FPSAC 2018).
19. Arithmetic dynamical Mordell–Lang, and Noncommutative arithmetic dynamical Mordell–Lang, with M. Zieve, Chapters 5–6 of the author’s dissertation, University of Michigan (2019), [available on Deep Blue](#).

TEACHING

Vassar College

- Math 261, Introduction to Number Theory: Fall 2023 (2 sections), Fall 2024 (2 sections), Spring 2025, Fall 2025 (2 sections)

- Math 361, Modern Algebra: Spring 2024, Fall 2024
- Math 331, Topics in Geometry: Fall 2025
- Math 301, Senior Seminar: Formalizing Mathematics (Spring 2025); Representation Theory (Fall 2026)
- Math 221, Linear Algebra: Spring 2024, Fall 2026 (2 sections)
- Math 399, Senior Independent Work: Galois Theory (3 students, Fall 2024); Advanced Algebra (1 student, Spring 2025); Category Theory (3 students, Fall 2025); Algebraic Number Theory (3 students), Computational Algebraic Geometry (1 student), and Topos Theory (1 student), Fall 2026
- On sabbatical leave, Spring 2026

University of Chicago

- Math 161–162, Honors Calculus (inquiry-based learning): 2019–2020
- Math 202, Abstract Linear Algebra: Winter 2020
- Math 242, Algebraic Number Theory: Spring 2022
- Math 254, Group Theory: Winter 2022
- Math 255, Rings and Fields: Winter 2022, Spring 2023

University of Michigan

- Math 116, Intermediate Calculus: seven semesters, 2014–2018; course co-coordinator, Winter 2018 (exams and assignments for 800+ students, 30+ instructors)
- Math 115, Introduction to Calculus: Fall 2013
- Math 215, Multivariable Calculus (lab instructor): Fall 2016

STUDENT RESEARCH AND MENTORING

URSI Mentor (Undergraduate Research Summer Institute)	Vassar College	2025
Directed Reading Program Mentor	University of Chicago	2021–2023
REU Mentor	University of Chicago	2020–2022
REU Mentor, Summer@ICERM (led to publications 7 and 9 above)	ICERM	2019
Math Circle Instructor	University of Michigan	2019
REU Mentor	University of Michigan	2015
Michigan Math and Science Scholars Instructor	University of Michigan	2015
Ross Mathematics Program Counselor	Ohio State University	2013

SERVICE AND ORGANIZING

Co-organizer, AMS Special Session on Arithmetic Dynamics	Joint Math Meetings	January 2027
Fellowship Committee	Vassar College	Fall 2026
Library Committee	Vassar College	Fall 2025
Co-organizer, Asprey Lecture Series	Vassar College	2023–present
Co-organizer, Department Colloquium (15+ speakers hosted)	Vassar College	2023–present
Speaker, DUSO lecture series	Vassar College	2024
Organizer, <i>Let's Talk About Math Grad School</i> information session	Vassar College	2023
Co-organizer, <i>No Boundaries Seminar</i>	University of Chicago	2021–2023
Organizer, <i>Farb and Friends Seminar</i>	University of Chicago	2021
Co-organizer, AMS Special Session <i>Current Trends in Arithmetic Dynamics</i>	Joint Math Meetings	2021
Organizer, Student Representation Theory Seminar	University of Michigan	2018

Referee for Proc. Amer. Math. Soc., Trans. Amer. Math. Soc., J. Théor. Nombres Bordeaux, New York J. Math., Finite Fields Appl., Int. J. Number Theory, Exp. Math., Proc. Roy. Soc. Edinburgh Sect. A, J. Combin. Theory Ser. A, Ergodic Theory Dynam. Systems, Int. Math. Res. Not. IMRN, Amer. Math. Monthly, Math. Intelligencer
Grant reviewer for the Israel Science Foundation (ISF) and the Austrian Science Fund (FWF)

INVITED TALKS

- Math Day Keynote Address, Colorado State University, November 2026 (*upcoming*)
- Number Theory Seminar, Academia Sinica, Taiwan, May 2026
- Algebra Seminar, National Central University, Taiwan, May 2026
- AG@PUI Seminar (online), March 2026
- Special Session on Arithmetic Dynamics, AMS Sectional Meeting, Boston College, March 2026
- Colloquium, Amherst College, March 2026
- AIM Workshop *Dynamics of Multiple Maps*, Caltech, November 2025
- Algebra/Number Theory Seminar, University of Rochester, March 2025
- Algebra Seminar, Florida State University, February 2025
- Special Session on Arithmetic Dynamics, Joint Mathematics Meetings, January 2025
- Algebra and Number Theory Seminar, Penn State University, November 2024
- Special Session on Arithmetic Dynamics, Joint Mathematics Meetings, January 2023
- Special Session on Arithmetic Dynamics, AMS Sectional Meeting, University of Utah, October 2022
- Arithmetic Dynamics International Online Seminar (ADIOS), February 2022
- Algebraic Dynamics Seminar, Harvard University, February 2022
- Number Theory Seminar, Oklahoma State University, March 2021
- Algebraic Geometry Seminar, University of Washington, February 2021
- Math Club, University of Connecticut, October 2020
- Special Session on Arithmetic Dynamics, Joint Mathematics Meetings, January 2020
- Combinatorial and Additive Number Theory Workshop, CUNY Graduate Center, May 2019
- Midwest Representation Stability Research Meeting, University of Chicago, April 2019
- Emerging Connections in Number Theory, AMS Sectional Meeting, March 2019
- Graduate Recruitment Symposium, University of Michigan, March 2019
- Complex Analysis, Dynamics, and Geometry Seminar, University of Michigan, March 2019
- Combinatorics Seminar, University of Michigan, February 2019
- Algebra Seminar, Wayne State University, January 2019
- Combinatorics Seminar, University of California, Berkeley, November 2018
- Geometry/Topology Seminar, University of Chicago, November 2018
- Formal Power Series and Algebraic Combinatorics (FPSAC), Dartmouth College, July 2018
- Combinatorial and Additive Number Theory Workshop, CUNY Graduate Center, May 2018
- Special Session on Arithmetic Dynamics, AMS Sectional Meeting, Northeastern University, April 2018
- Special Session on Arithmetic Dynamics, Joint Mathematics Meetings, January 2018
- Interactions between Model Theory and Arithmetic Dynamics, Fields Institute, July 2016
- Colloquium, Amherst College, April 2013
- Pi Day Colloquium, University of Connecticut, March 2013